

HAIFA ECO PULSE

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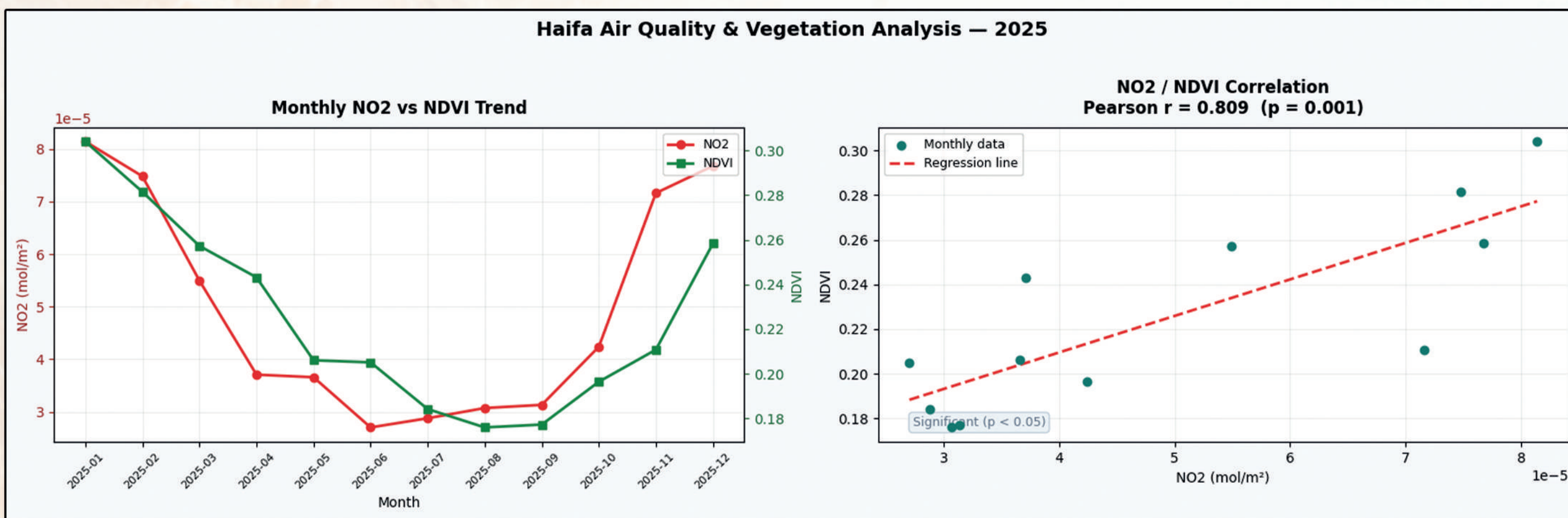
Background & Motivation

Industrial pollution in Haifa Bay threatens the ecosystem, yet existing tools lack the automated statistical engines needed to quantify this damage.

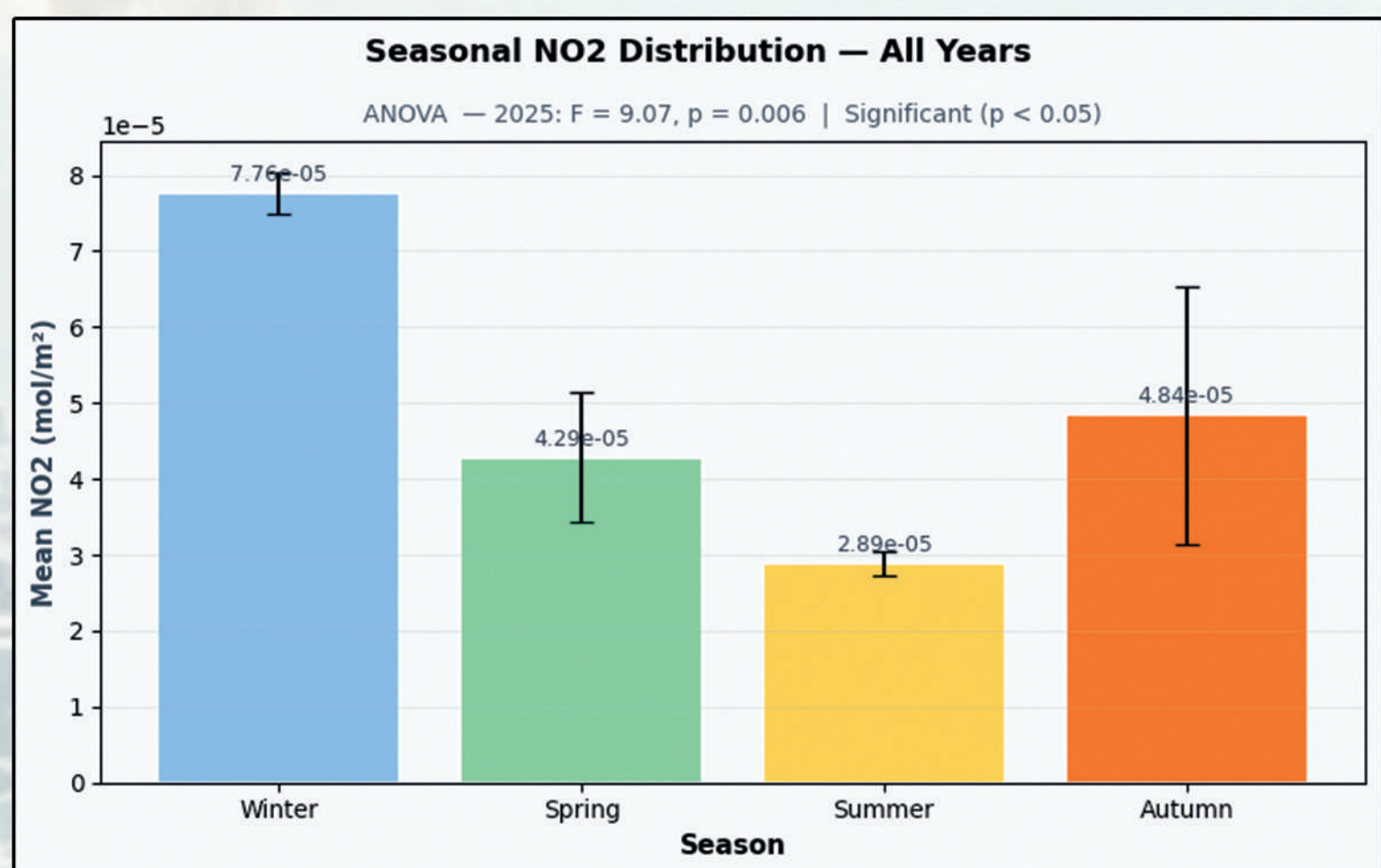
Haifa Eco Pulse bridges this gap by integrating Google Earth Engine satellite data (Sentinel-5P NO_2 and Sentinel-2 NDVI) to track the correlation between air pollution and vegetation health.

Analytics & Statistical Testing

Pearson Correlation: Utilized for RQ1 to check the strength and direction of the linear relationship between two continuous variables (gas levels and vegetation index).

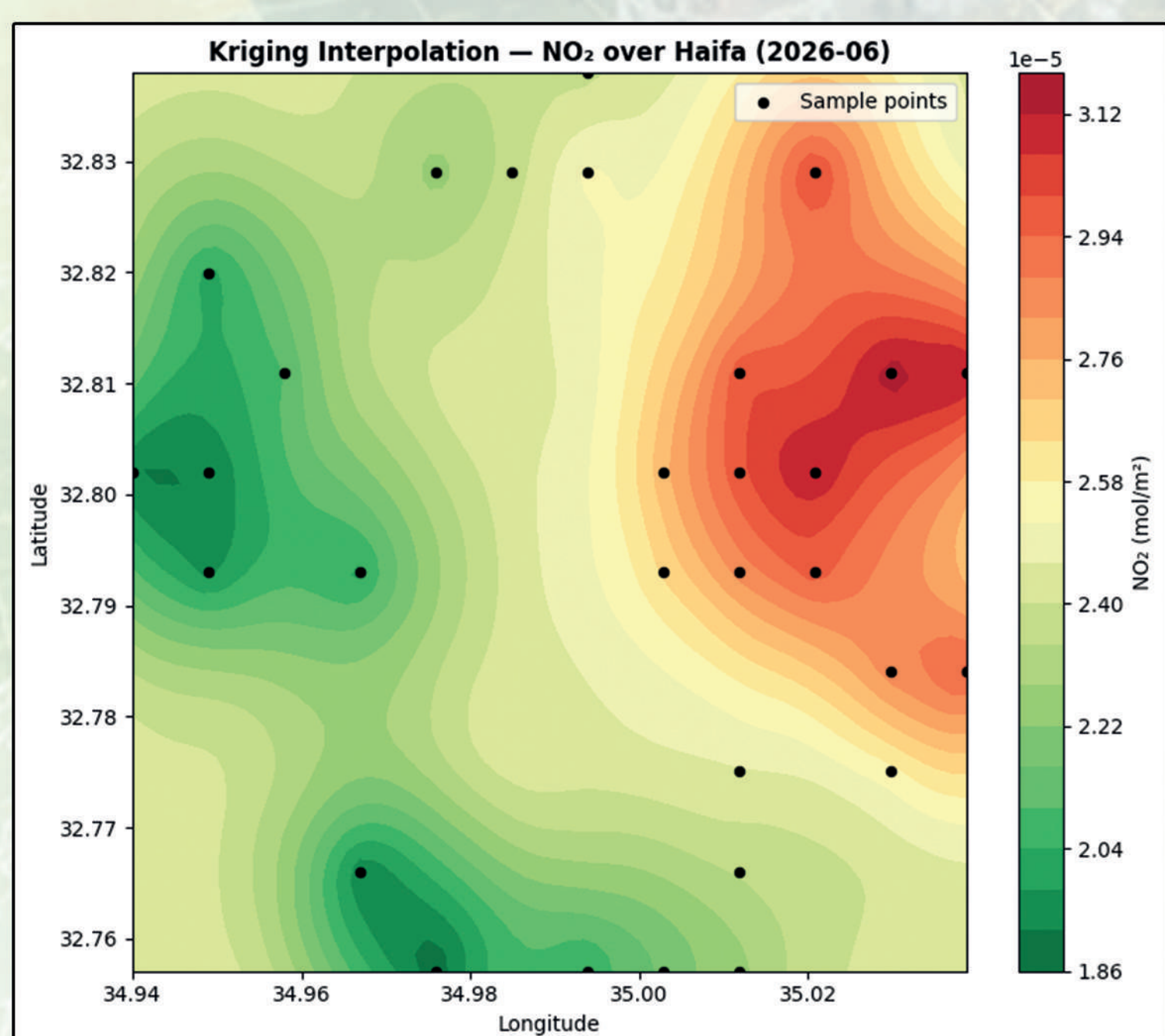


One-Way ANOVA: Utilized for RQ2 to measure the average air pollution level and determine if there is a difference in means between the four seasons. If a difference is observed, a **Tukey's HSD** test will be performed to compare the summer months against each of the other seasons separately.



Ordinary Kriging: Satellite images often suffer from missing data due to cloud cover.

To solve this, we use an **Ordinary Kriging** model to fill in the gaps, transforming scattered pixel samples into a smooth, continuous 60x60 heat map.



Research Questions & Hypotheses

RQ1: Is there a significant correlation between air pollution levels (NO_2) and vegetation health (NDVI) in Haifa?

H1: A significant negative correlation exists ($r < 0$) higher NO_2 levels correspond to lower NDVI (degraded vegetation).

H0: There is no correlation between NO_2 levels and NDVI ($r = 0$).

RQ2: Do air pollution levels (NO_2) in Haifa change significantly between the four seasons of the year, and are the summer months the most polluted compared to each season separately?

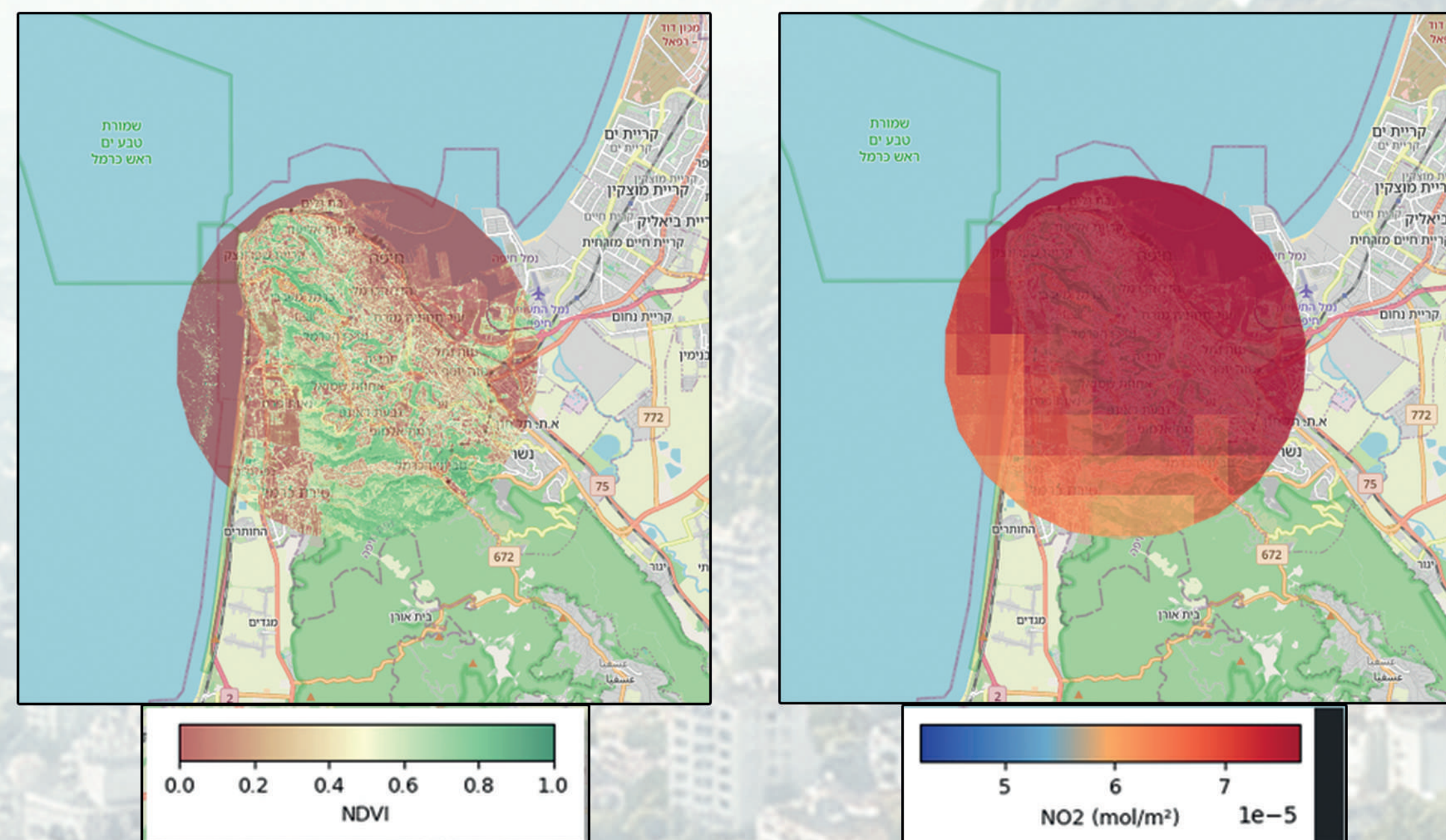
H1: NO_2 concentrations differ significantly by season, with distinct peaks expected during the summer months.

H0: Mean NO_2 concentrations are equal across all four seasons.

Interactive Spatio - Temporal Map

Instead of just looking at static data, users can use a time slider to scroll through monthly satellite records.

By layering NO_2 pollution footprints directly over NDVI vegetation maps, it makes it easy to visually track how emissions spread across Haifa's landscape and impact the ecosystem month by month.



Discussion & Key Learnings

Temporal Correlation (RQ1): **Pearson** analysis across multiple annual datasets demonstrates significant variance, revealing that the relationship between NO_2 and NDVI is not consistently inverse.

Seasonal Variance (RQ2): **One-Way ANOVA** consistently confirms a statistically significant difference in mean NO_2 concentrations across seasons.

Across all analyzed periods, the data contradicts the initial summer-peak hypothesis. Winter consistently records the maximum mean pollution levels, while Summer records the absolute minimum.